Assignment 2: Build Your Own Tool

Due Week 3 at the start of class.

Goal Build and deploy a multi-page Next.js + Tailwind app that you'd actually use.

Overview

Build something you'd actually open tomorrow. A tool for your life — planning, tracking, organizing, journaling. The app stores data in client-side state — it lives in memory and disappears on refresh. That's the point. You'll feel the limitation, and next week we fix it with a database. For now: exercise the stack — routing, components, forms, Tailwind, and the build → verify loop with Playwright MCP.

Ideas to Explore (Pick One)

📅 Day Planner — tasks, notes, and schedule per day. Week overview. Navigate between days.
📚 Reading Log — books with status (reading / finished / want to read), notes, ratings, a shelf view.
💪 Workout Tracker — exercises per day, sets and reps, weekly summary, personal records page.
🍳 Recipe Book — recipes with ingredients, steps, tags. Search and favorites.
💼 Interview Prep — Practice questions, company research notes, a preparation checklist, and a progress tracker.
🎬 Watchlist — movies and shows with status, ratings, reviews, a "recommend" view.
📝 Course Notes — organized by week or topic, searchable, with linked references.
🧠 Flashcard Maker — decks, cards, a study mode page, progress tracking.

✨ Invent Your Own — The pattern is the same: multiple pages, a form to add data, a way to view it, and design iteration.

Requirements

(If any of this is unclear—ask Claude for help!)

- At least 4 distinct pages / routes
- A form that adds data (client-side state — data lives in memory for now)
- A dynamic route (e.g., /day/[date] or /recipe/[slug])
- A shared layout with navigation between pages
- Styled with Tailwind — it should look like something you'd show someone
- A CLAUDE.md that describes your project, pages, and data model
- Playwright MCP configured — ask Claude to verify at least one interaction
- Deployed to Vercel with a live URL
- Code pushed to a public GitHub repo
- Multiple git commits showing your iteration process

Steps

1. **Pick your tool**. What would you actually use? Don't overthink it — pick something and commit.
2. **Plan with Claude**. Put Claude in Plan Mode. Describe the app, the pages, the data model, the style. Have Claude interview you on what you want to build and steer it in the right direction.
3. **Scaffold with Claude**. Create the Next.js app, set up Tailwind, get the dev server running.
4. **Configure Playwright MCP**. Ask Claude to open your app and verify something works — a form submission, a navigation link, a page layout.
5. **Build page by page**. Start with the homepage. Get it looking good. Then add the form page. Then the dynamic route. Then the overview.
6. I**terate on the design and interactions**. This is where the assignment gets interesting. Claude can scaffold fast — but can you direct it to make something polished? Push on typography, spacing, color, layout. Explore animations.
7. **Deploy to Vercel**. Push to GitHub, connect to Vercel, ship it. Note: your data lives in memory — it won't survive a page refresh. That's expected — and it's why Week 3 adds a database.

Submission

Deliverables

1. Vercel URL — A live link to your tool.

2. GitHub URL — Public repository with commit history.

3. Reflection Questions

1. What's in your CLAUDE.md? How did your plan shape what Claude built — and how did it evolve as you worked?

2. Exercise your mental model: Pick one page in your app. Trace the path: what file renders it, what's the route, what components does it use, where does the data come from?

3. Describe one thing that happened when Claude tested your app with Playwright MCP. How did the build → verify loop change how you worked?

Example Submission

```
https://my-site.vercel.app
https://github.com/username/my-site

1. <answer to question 1>
2. <answer to question 2>
3. <answer to question 3>
```

Assessment

A complete assignment has:

- ☐ A live Vercel URL that works
- ☐ A multi-page app that captures and displays data (at least 4 pages)
- ☐ A GitHub repo with multiple commits showing iteration
- ☐ Thoughtful answers to the 3 reflection questions

You'll be assessed on whether you used the tools, iterated with intention, and built something with depth

Notes

Commit and push as you go — it lets you rollback when needed. Having Claude do the commits speeds things up.

Explore Claude Code

CLAUDE.md — Create one in your project root with what you're building and style preferences.

Plan mode — Shift+Tab to plan before building. Read the plan, then let Claude build.

Prompting — Try vague vs. specific prompts. Iterate: "bolder typography," "more whitespace."

Screenshots — Paste a screenshot of a site you love and say "make something inspired by this."

Commands — /context · /compact · /clear · /diff · Esc Esc (undo) - https://code.claude.com/docs/en/commands

Questions? Reach out on Slack.